

Modelling for outbreak response: predictions, evaluation, and utility

4th MONID Conference, Halle (Saale), 21 May 2026



Sebastian Funk

<https://epiforecasts.io>

LONDON
SCHOOL *of*
HYGIENE
& TROPICAL
MEDICINE



**centre *for*
mathematical
modelling *of*
infectious diseases**

A hantavirus outbreak on a cruise ship



Photo: Stefan Brending

A policy question

How long does the quarantine period have to be?

Epuyén, Argentina, 2018–19

A cluster of 34 cases of Andes hantavirus, with sustained person-to-person transmission.

Incubation periods ranged from 9 to 40 days

Martínez et al., *NEJM*, 2020

Re-estimating the incubation period

LogNormal distribution, posterior median (95% CrI):

days	
Mean	22.6 (20.2 – 25.5)
95th percentile	36.2 (31.3 – 44.5)
99th percentile	45.0 (37.6 – 58.7)

<https://github.com/epiforecasts/andv-linelist-analysis>

"All models are wrong, but some are useful."

George Box

- Was our model **right**? was it **useful**?

Challenges of modelling in real time

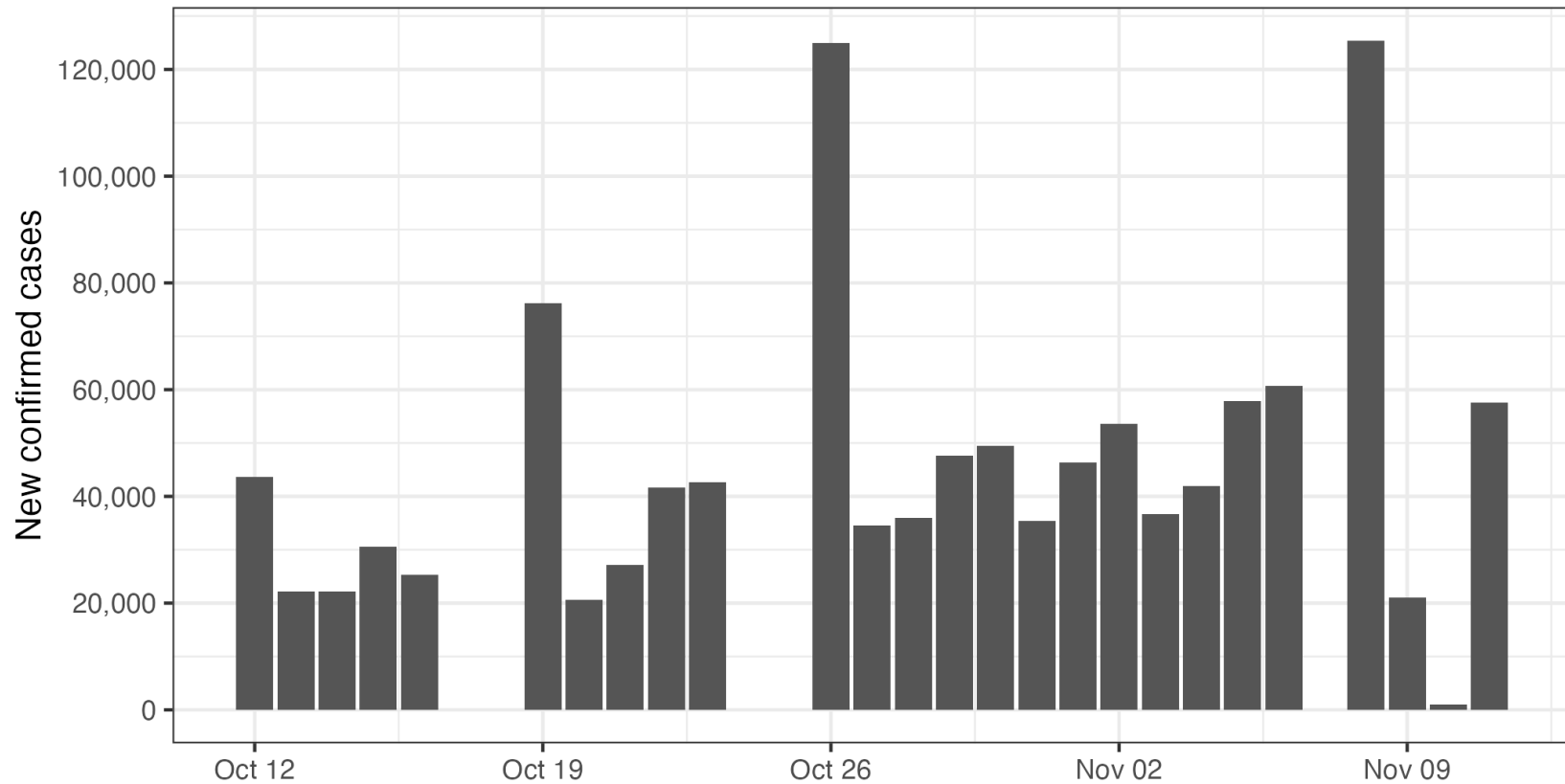
"We were losing ourselves in details [...] all we needed to know is, are the number of cases rising, falling or levelling off?"

Hans Rosling, Liberia, 2014

or: what is the current value of the reproduction number R_t

The answer to the question is not always obvious

COVID-19 cases in France, as of 12 November 2020



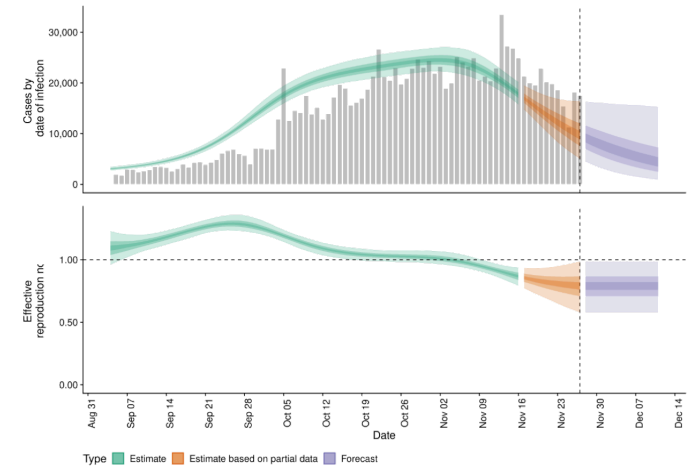
A generative Bayesian model for estimating R_t

$$\text{New infections } I(t) = R_t \sum_{\tau} g_{\tau} I_{t-\tau}$$

$$\text{Reproduction number } R(t) = R_{t-1} \exp(\text{GP})$$

$$\text{Delayed reporting } D(t) = \sum_{\tau} \xi_{\tau} I_{t-\tau}$$

$$\text{Observations } C(t) = \text{NegBin}(D_t \omega_{(t \bmod 7)}, \phi)$$



implemented in the [EpiNow2](#) R package.

January 2020

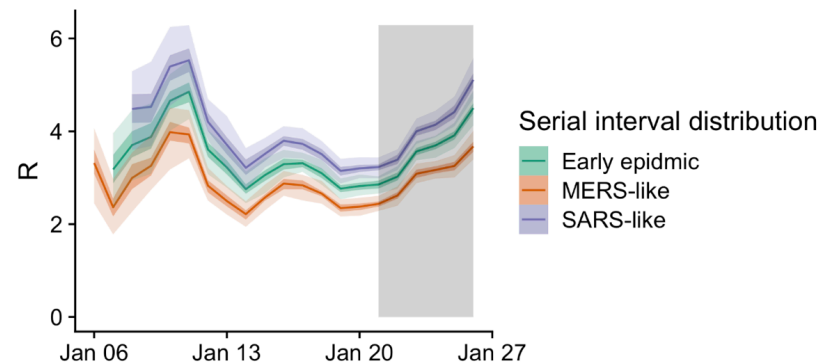
Reporting delays and temporal variation in transmission in China during the 2019-nCoV outbreak

Status: In Progress | First online: 30-01-2020 | Last update: 30-01-2019

Authors: [Sebastian Funk*](#), Sam Abbott, Stefan Flasche & [CMMID COVID-19 working group](#).

Time-varying reproduction number

In the time-varying reproduction number, we see differences depending on the serial interval distribution used. In the early epidemic the latest estimate of the basic reproduction number is 5. There is no apparent downward trend in the R uptick in the reproduction number. This may be an artifact of the adjustment to confirmation delays, especially if assumed here.



Funk et al., cmmid.github.io, 2020

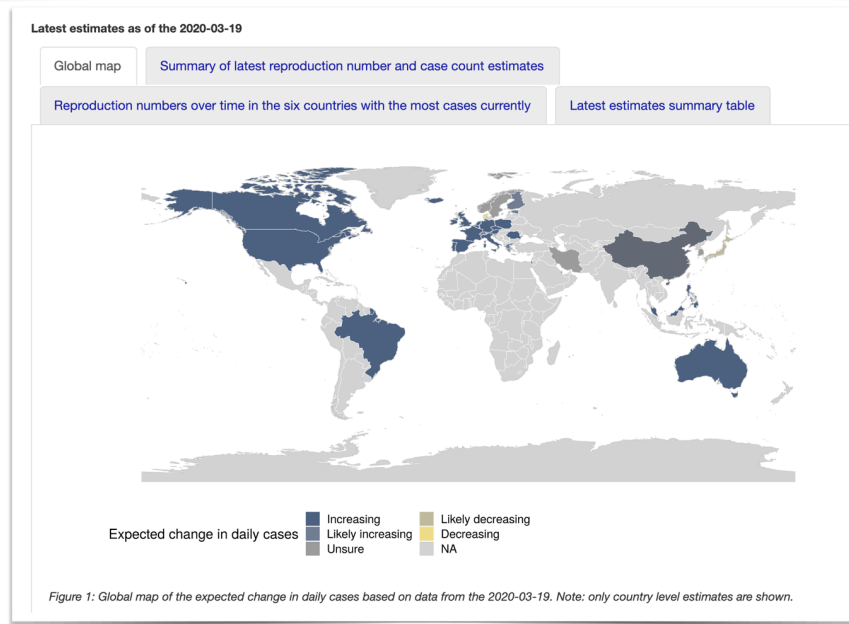
March 2020

Temporal variation in transmission during the COVID-19 outbreak

Status: In Progress | First online: 02-03-2020 | Last update: 04-04-2020

Authors: [Sam Abbott*](#), Joel Hellewell, James D Munday, June Young Chun, Robin N. Thompson, Nikos I Bosse, Yung-Wai Desmond Chan, Timothy W Russell, Christopher I Jarvis, [CMMID COVID-19 working group](#), Stefan Flasche, Adam J Kucharski, Rosalind M Eggo & Sebastian Funk.

* corresponding author



[Abbott et al., *cmmid.github.io*, 2020](#)

August 2020

Global summary

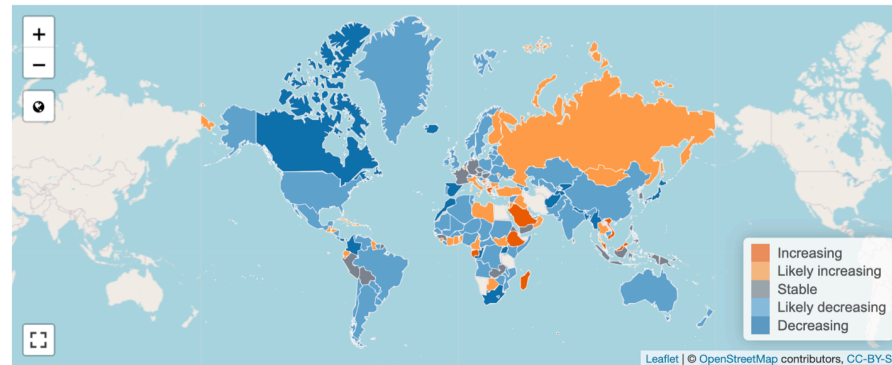
GLOBAL SUMMARY

Identifying changes in the reproduction number, rate of spread, and doubling time during the course of the COVID-19 outbreak whilst accounting for potential biases due to delays in case reporting both nationally and subnationally. These results are impacted by changes in testing effort, increases and decreases in testing effort will increase and decrease reproduction number estimates respectively.

Using data available up to the: 2021-02-01

Estimates (based on reported cases and deaths) are available for download [here](#).

See our [see Methods](#) or our [paper](#) for an explanation of how these estimates are derived.



<https://epiforecasts.io/covid/posts/global/>

Subnational estimates

Subnational breakdown

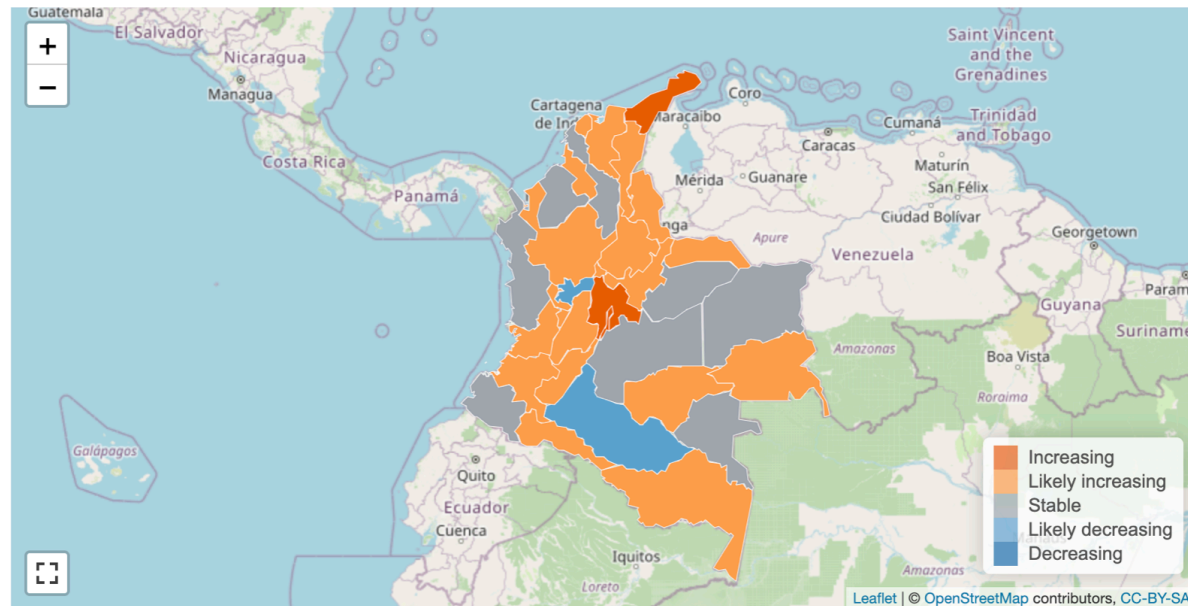


Figure 2: The results of the latest reproduction number estimates (based on confirmed cases in Colombia, stratified by Region, can be summarised by whether confirmed cases are likely increasing or decreasing. This represents the strength of the evidence that the reproduction number in each region is greater than or less than 1, respectively (see the [methods](#) for details).

CSV

Search:

Daily estimates stored on github

 README

 Code of conduct

 MIT license





National and subnational estimates of the time-varying reproduction number for Covid-19

License

MIT

Status

csv

website

up

This repository contains estimates of the time-varying reproduction number for every country in the world listed in the ECDC Covid-19 data source and subnational estimates for 9 countries.

Summarised estimates as csv's can be found in the `**/summary` folders. National estimates can be found in the `national` folder and subnational estimates in the `subnational/country` subfolder depending on the country of interest.

All regions have R_t estimates based on case counts. In some regions, we also separately estimate R_t using counts of hospital admissions or deaths. The differences between each of these estimates might suggest uneven or changing transmission by age and/or severity in the general population. For more on the influence and different uses of data source on R_t , see [here](#).

Latest results

Estimates are generated using `{EpiNow2}` and presented on epiforecasts.io/covid (which also outlines the method used). If using these estimates please consider citing our associated [paper](#).

Was the model right?

Some methodological challenges

Challenge #1: methodological flaws

Originally used methodology that ignored delays.

EpiNow0.3.0

HomeGetting startedNewsFunctions

EpiNow: Estimate realtime case counts and time-varying epidemiological parameters

{EpiNow} has been deprecated in favour of {EpiNow2}

This package estimates the time-varying reproduction number, rate of spread, and doubling time using a range of open-source tools and current best practices. It aims to help users avoid some of the limitations of naive implementations in a framework that is informed by community feedback and is under active development. It assumes that only limited data is available on cases by date of onset and instead uses cases by date of report. These are then imputed to case counts by date of infection using an uncertain reporting delay and incubation period. Right truncation of cases is dealt with internally by `{EpiNow}`, as is propagating uncertainty from all inputs into the final parameter estimates (helping to mitigate spurious findings). Time-varying estimates of the reproduction number are estimated using the `{EpiEstim}` package by date of infection with a generation time estimate that includes uncertainty. Time-varying estimates of the rate of growth are derived using a quasipoisson GLM with a sliding window, which are then used to estimate the doubling time. Optimal windows are chosen by using one day ahead case prediction. Optionally, the time-varying reproduction number can be forecast forwards in time using an integration with the `{EpiSoon}` package and converted to a case forecast using a branching process. See the [methods](#) section of our Covid-19 site for a detailed discussion of the approach.

PERSPECTIVE

Practical considerations for measuring the effective reproductive number, R_t

Katelyn M. Gostic^{1*}, Lauren McGough¹, Edward B. Baskerville¹, Sam Abbott², Keya Joshi³, Christine Tedijanto³, Rebecca Kahn³, Rene Niehus³, James A. Hay³, Pablo M. De Salazar³, Joel Hellewell², Sophie Meakin², James D. Munday², Nikos I. Bosse², Katharine Sherratt², Robin N. Thompson^{2,4}, Laura F. White⁵, Jana S. Huisman^{6,7}, Jérémie Scire^{7,8}, Sebastian Bonhoeffer⁶, Tanja Stadler^{7,8}, Jacco Wallinga^{9,10}, Sebastian Funk², Marc Lipsitch³, Sarah Cobey¹

Gostic et al., *PLOS Comp Biol*, 2020

Challenge #2: lack of local context

This results in an onset to case report delay distribution with a mean of 6.5 days and a standard deviation of 17 days and an onset to death report delay distribution with a mean of 13.1 days and a standard deviation of 11.7 days. For computational reasons the maximum allowed delay is set to be 30 days. Dataset specific estimates are detailed on the respective country pages. Estimated delays are routinely updated as new data becomes available.

As data may also be right truncated due to unrecorded delays (i.e the delay between a case report and its appearance in an aggregated data set) we truncate all time-series to exclude the last 3 days of data, based on qualitative inspection of the stability of case counts in the datasets used.

We used an estimate of the generation time sourced from¹⁸ but refit using a log-normal incubation period with a mean of 5.2 days (SD 1.1) and SD of 1.52 days (SD 1.1)¹⁹ rather than the incubation period used in the original study (code available here: <https://github.com/seabbs/COVID19>). This resulted in a distributed generation time with mean 3.6 days (standard deviation (SD) 0.7), and SD of 3.1 days (SD 0.8) for all estimates. The incubation period estimate was also used to convolve from unobserved infections to unobserved onsets in the model. See 10 for further details on the approach.

Abbott et al., Wellcome Open Res, 2020

Challenge #3: Lack of capacity of manual curation

Once again, UNBELIEVABLE data. Iowa, in the United States, again!!!!
Seriously??? #189

✓ Closed

ghost opened this issue on Dec 13, 2021 · 1 comment



ghost commented on Dec 13, 2021

...

Once again, you have screwed up the data for Iowa in the United States. Your data shows more new cases in the state than there are residents.

Is ANYONE in your group going to actually decide to bounds check your data before you publish it???

Please!!! Fix your data, and for God's sake, check it before you post. Your site is unreliable and cannot be trusted.

Assign

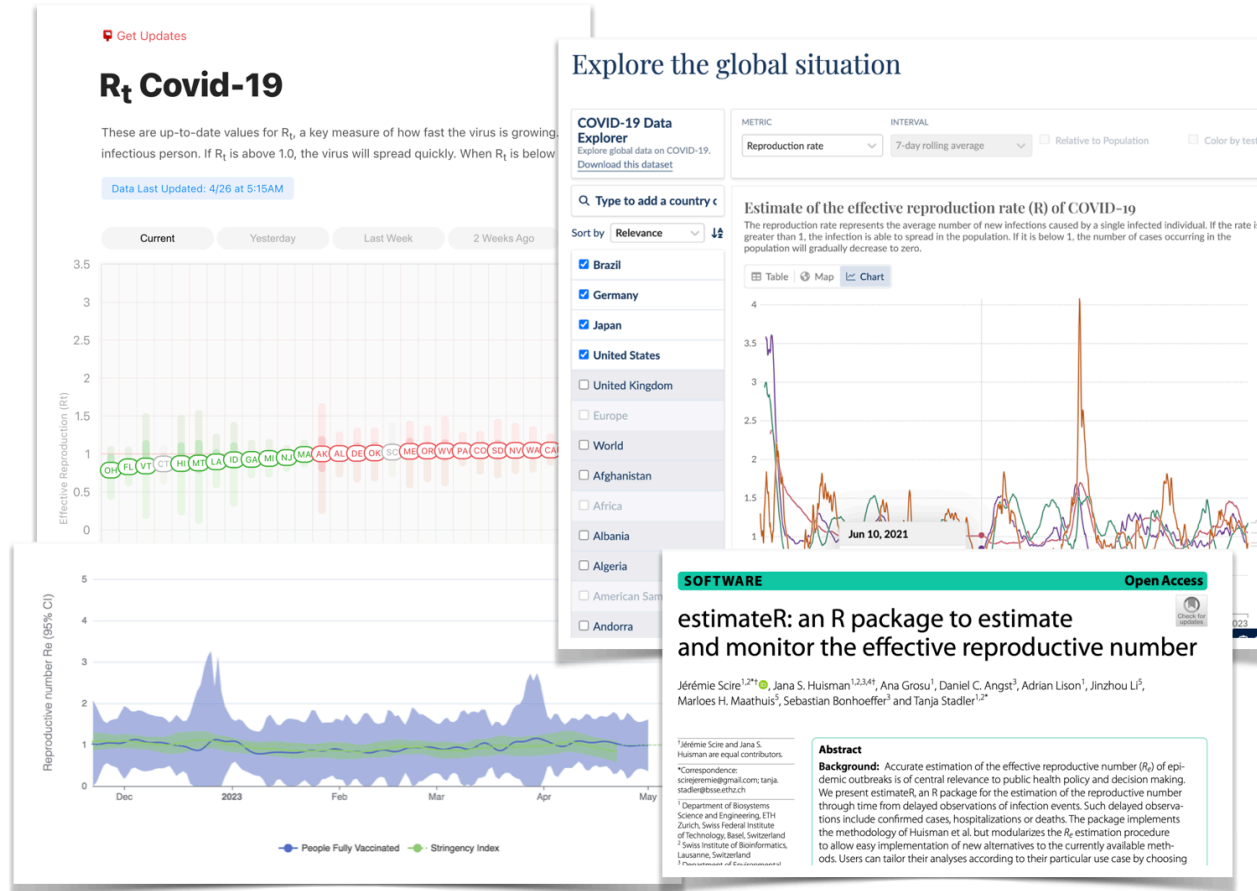
No one

Label

None

Was it useful?

Challenge #4: Multiplication of effort



Brockhaus et al., *PLOS Comp Biol*, 2023

Impact #1: Use by other researchers

Drivers of epidemic dynamics in real time from daily digital COVID-19 measurements

Michelle Kendall^{1,4*}, Luca Ferretti^{2,3,4*}, Chris Wymant^{2,3}, Daphne Tsallis⁴, James Petrie^{2,3}, Andrea Di Francia⁵, Francesco Di Lauro^{2,3}, Lucie Abeler-Dörner^{2,3}, Harrison Manley⁵, Jasmina Panovska-Griffiths^{2,3,5}, Alice Ledda³, Xavier Didelot^{1,6}, Christophe Fraser^{2,3,6}

Cite as: M. Kendall *et al.*, *Science* 10.1126/science.adm8103 (2024).

RESEARCH ARTICLE

Open Access

The impact of non-pharmaceutical interventions on SARS-CoV-2 transmission across 130 countries and territories

Yang Liu^{1†}, Christian Morgenstern^{2†}, James Kelly², Rachel Lowe^{1,3}, CMMID COVID-19 Working Group and Mark Jit^{1*}

PNAS

RESEARCH ARTICLE

EVOLUTION

OPEN ACCESS

Estimates of early outbreak-specific SARS-CoV-2 epidemiological parameters from genomic data

Timothy G. Vaughan^{a,b,1}, Jérémie Scire^{a,b}, Sarah A. Nadeau^{a,b}, and Tanja Stadler^{a,b,1}

Edited by Nils Stenseth, Universitetet i Oslo, Oslo, Norway; received May 15, 2023; accepted December 2, 2023

PLOS COMPUTATIONAL BIOLOGY

RESEARCH ARTICLE

Challenges of COVID-19 Case Forecasting in the US, 2020–2021

Velma K. Lopez^{1*}, Estee Y. Cramer², Robert Pagano³, John M. Drake⁴, Eamon B. O'Dea⁵, Madeline Adee⁶, Turgay Ayer⁷, Jagpreet Chhatwal⁷, Ozden O. Dalgic⁸, Mary A. Ladd⁹, Benjamin P. Linares¹⁰, Peter P. Mueller¹¹, Jade Xiao¹², Johannes Bracher¹³, Alvaro J. Castro Rivadeneira¹⁴, Aaron Gerding¹⁵, Tilmann Gnelting¹⁶, Yuxin Huang¹⁷, Dasuni Jayawardena¹⁸, Abdul H. Kanji¹⁹, Khoa Le²⁰, Anja Mühlemann¹², Jarad Niemi¹³, Evan L. Ray²¹, Ariane Stark²², Yijin Wang²³, Nutch Wattanachit²⁴, Martha W. Zorn²⁵, Sen Pei¹⁴, Jeffrey Shaman¹⁴, Teresa K. Yamana¹⁴, Samuel R. Tarasewicz¹⁵, Daniel J. Wilson¹⁵, Sid Baccam¹⁶, Heidi Gurung¹⁶, Steve Stage¹⁷, Brad Suchoski¹⁶, Lei Gao¹⁸, Zhiling Gu¹⁵, Myungjin Kim¹⁹, Xinyi Li²⁰, Guannan Wang²¹, Lily Wang²², Yueying Wang²³, Shan Yu²⁴, Lauren Gardner²⁵, Sonia Jindal²⁶, Maximilian Marshall²⁴, Kristen Nixon¹⁴, Juan Dent²⁷, Allison L. Hill²⁸, Joshua Kaminsky²⁹, Elizabeth C. Lee²⁵, Joseph C. Lemaitre²⁵, Justin Lessler²⁷, Claire P. Smith²⁸, Shaun Truelove²⁵, Matt Kinsey²⁸, Luke C. Mullany²⁸, Kaitlin Rainwater-Lovett²⁸, Lauren Shin²⁸

Abbott et al., medRxiv, 2021

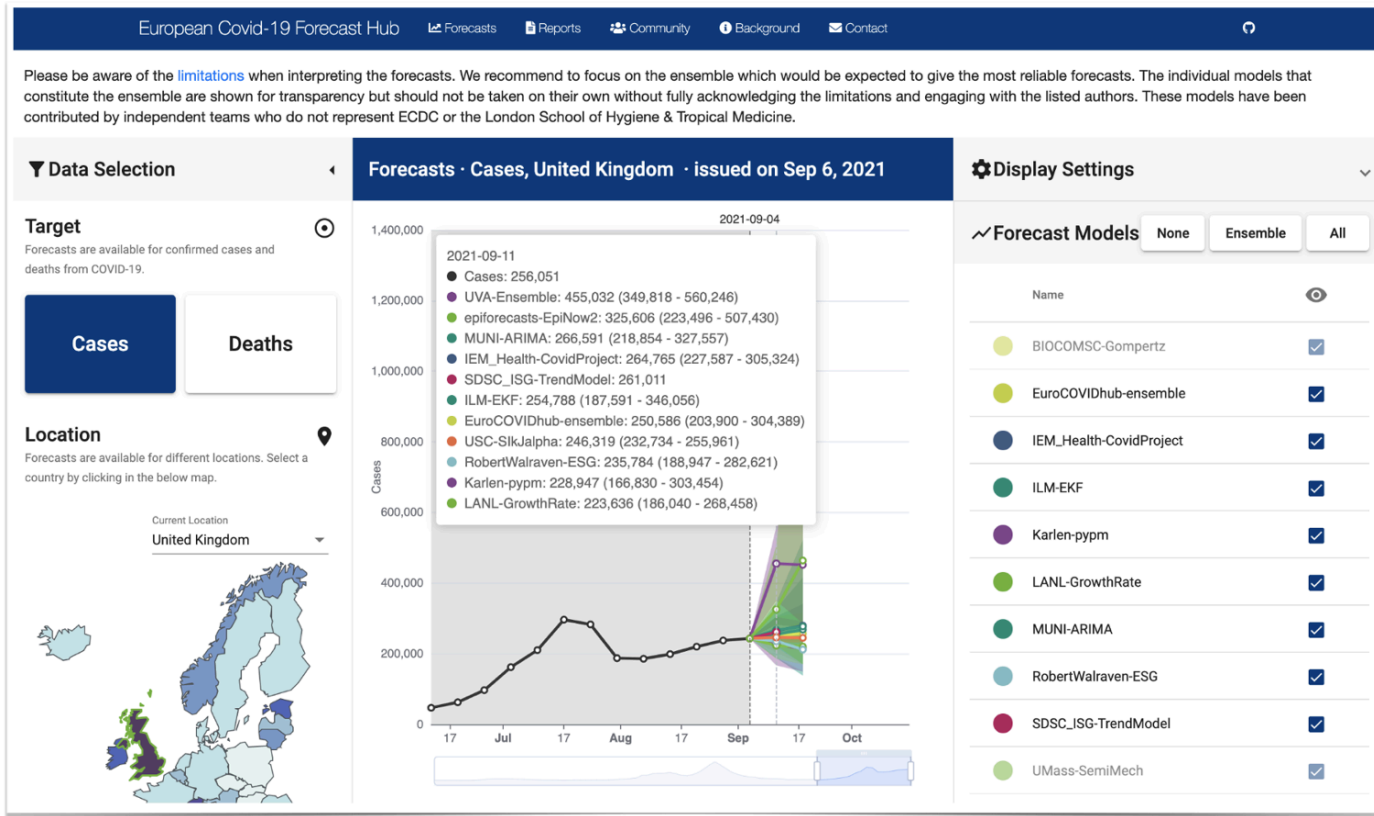
Sheratt et al., *Proc Roy Soc B*, 2021

Impact #3: Use in public health agencies

"The Rt estimates generated by the EpiForecasts team were used extensively by the WHO COVID-19 Analytics unit. They formed a core part of two routine analysis pipelines [...] regularly presented to the incident management structure at WHO headquarters, including senior management, as well as regional and national WHO offices."

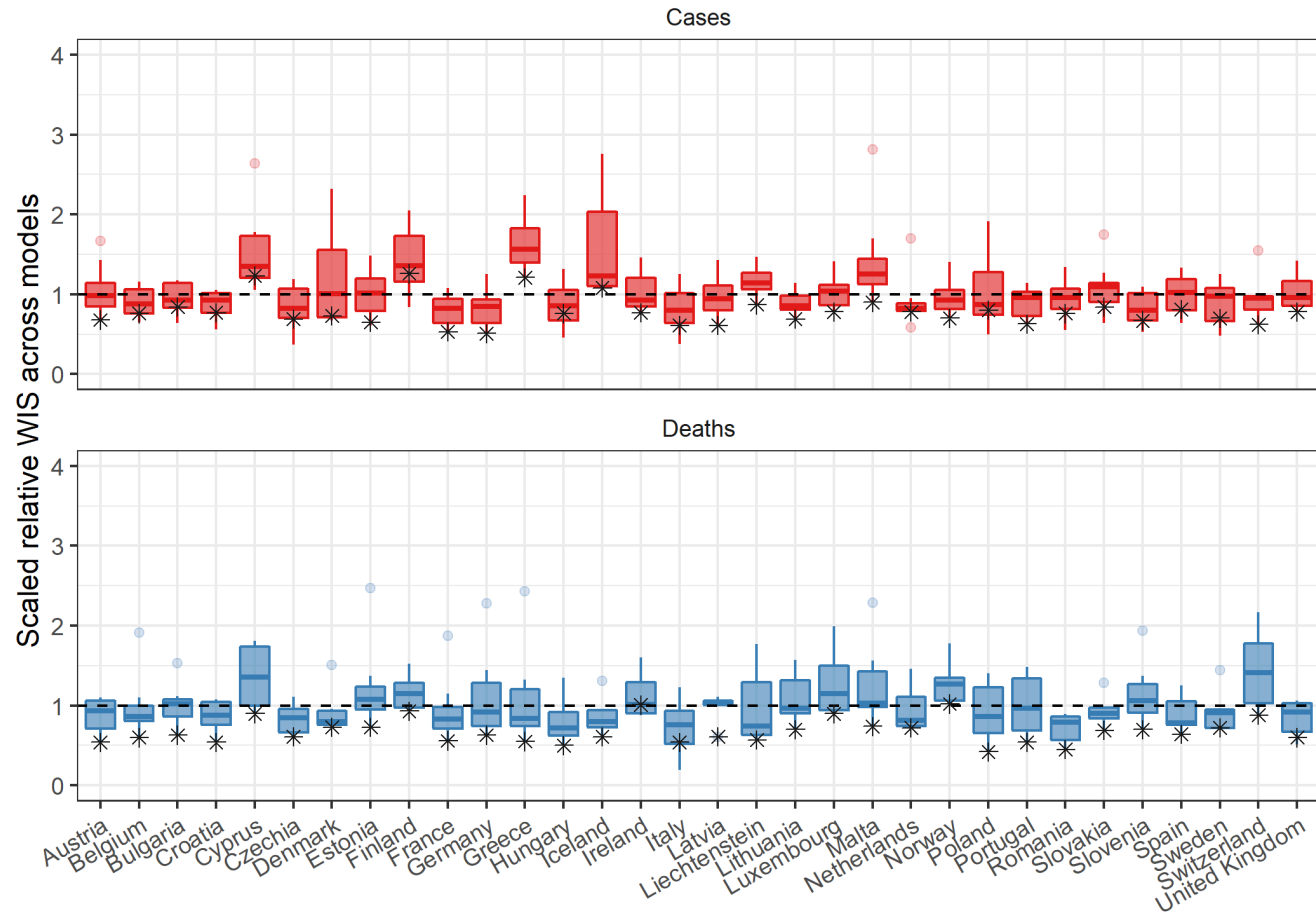
In order to tell how wrong we are, we need
predictions confronted with reality

Forecast hubs support systematic collection and evaluation of forecasts



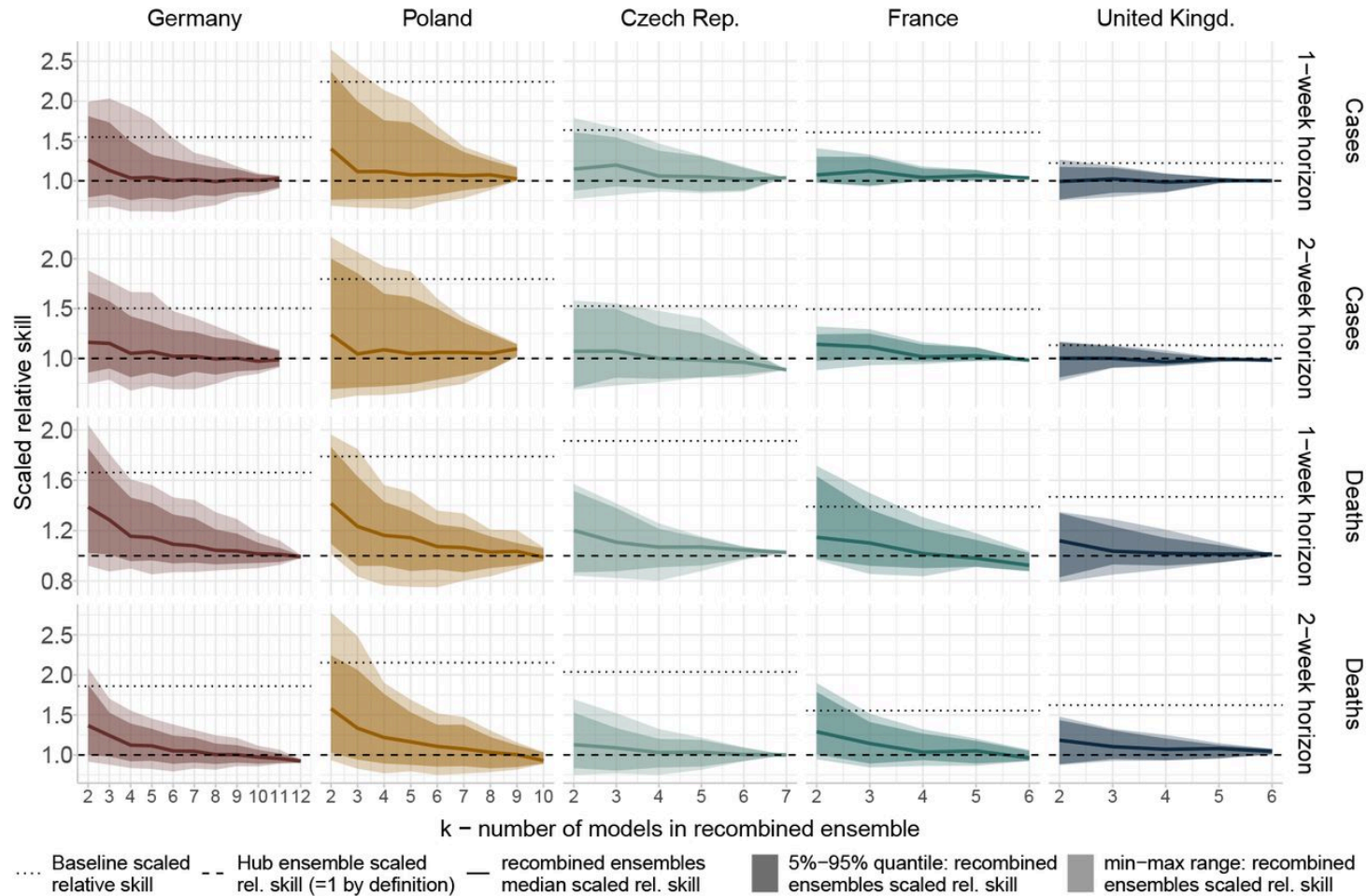
Reich et al., *Am J Public Health*, 2022

Ensembles beat individual models



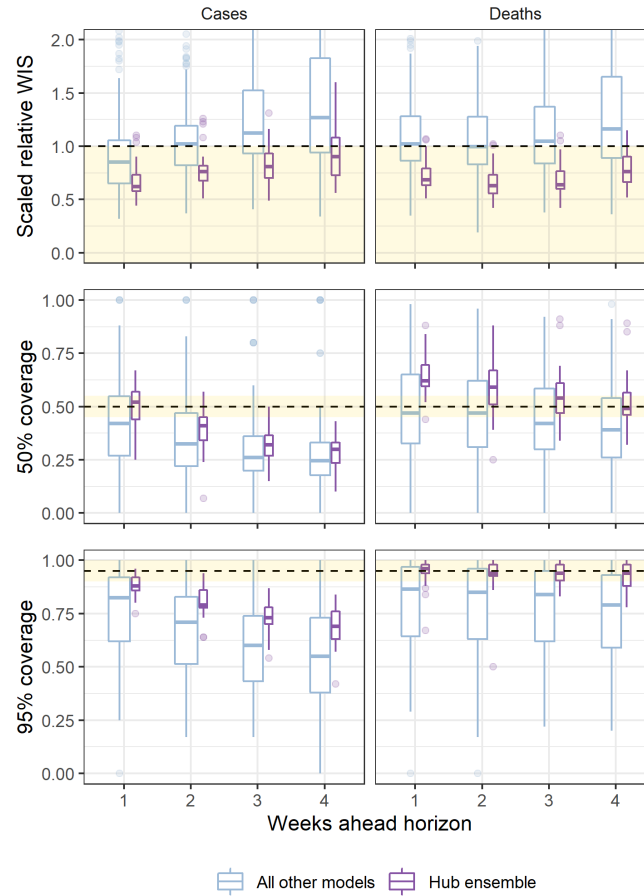
Sherratt et al., *eLife*, 2023

The more models in the ensemble, the better



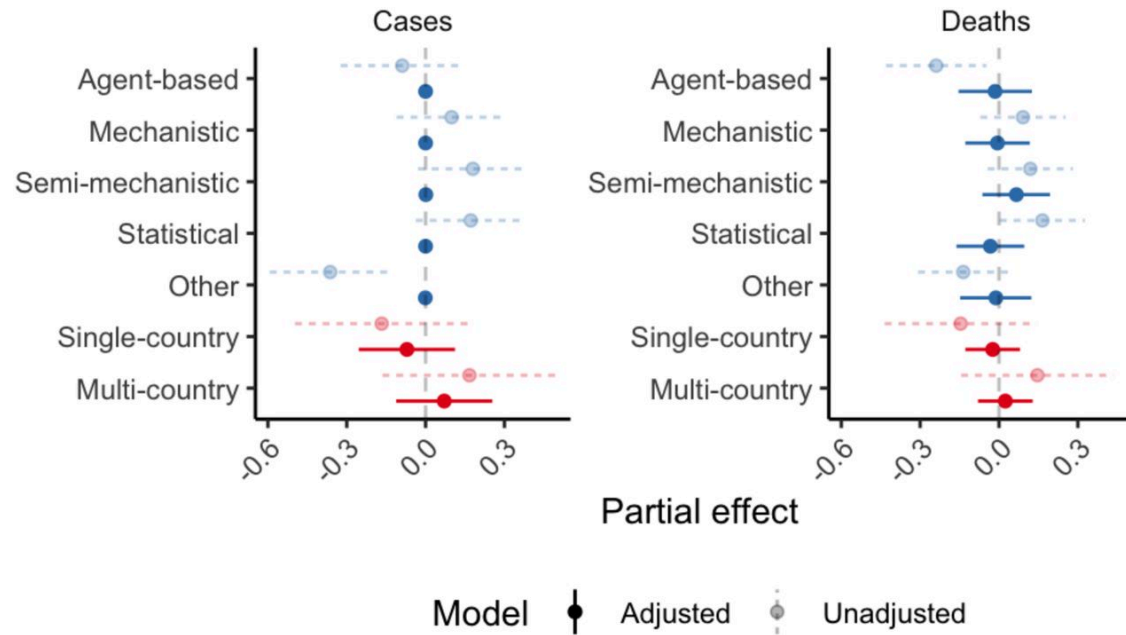
Becker et al., *medRxiv*, 2025 Fox et al., *EID*, 2024

Case forecasts degrade quickly



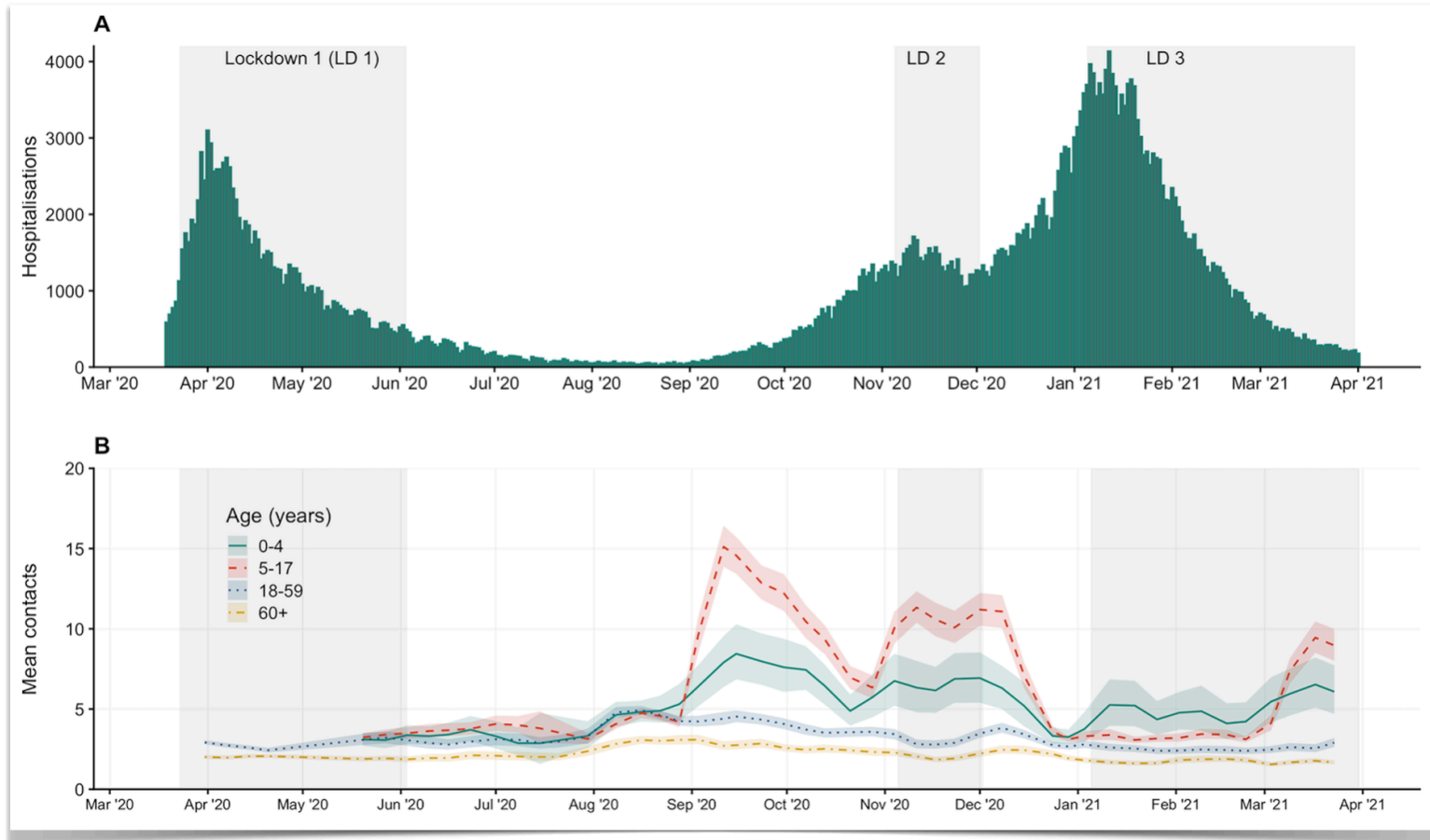
Sherratt et al., *eLife*, 2023

No model type dominates



Sherratt et al., *medRxiv*, 2025

Changes in behaviour are hard to predict



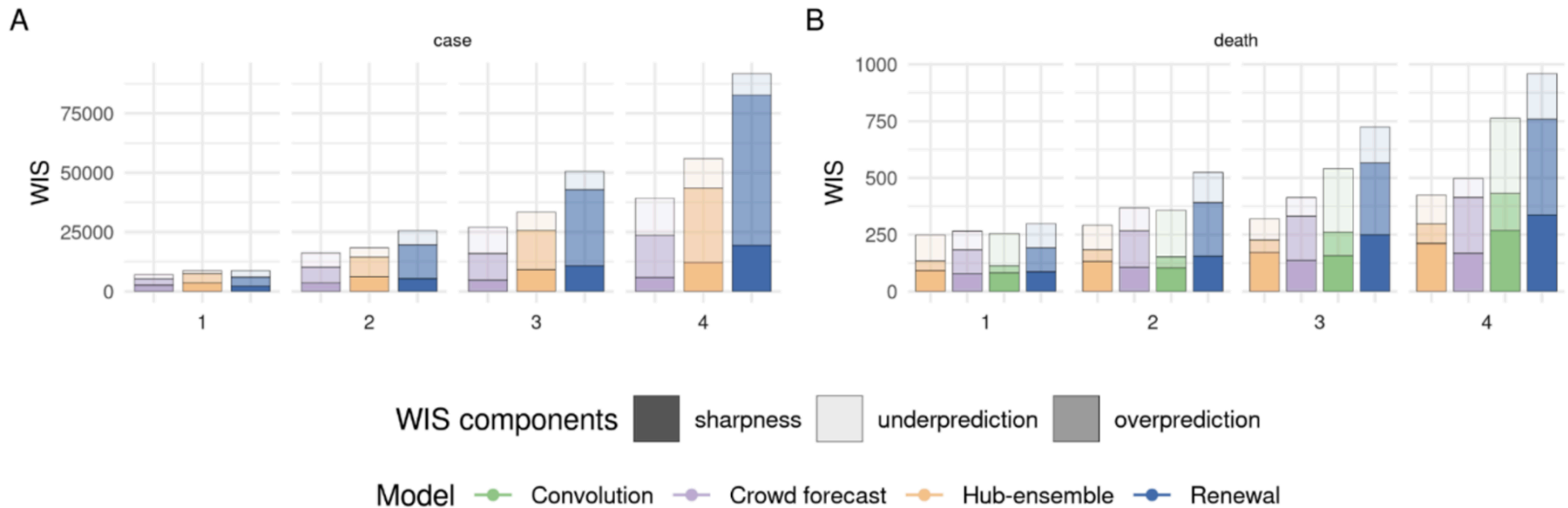
Gimma et al., *PLOS Medicine*, 2022

Comparing human forecasts to models



Bosse et al., *PLOS Comp Biol*, 2022
Bosse et al., *Wellcome Open Res*, 2024

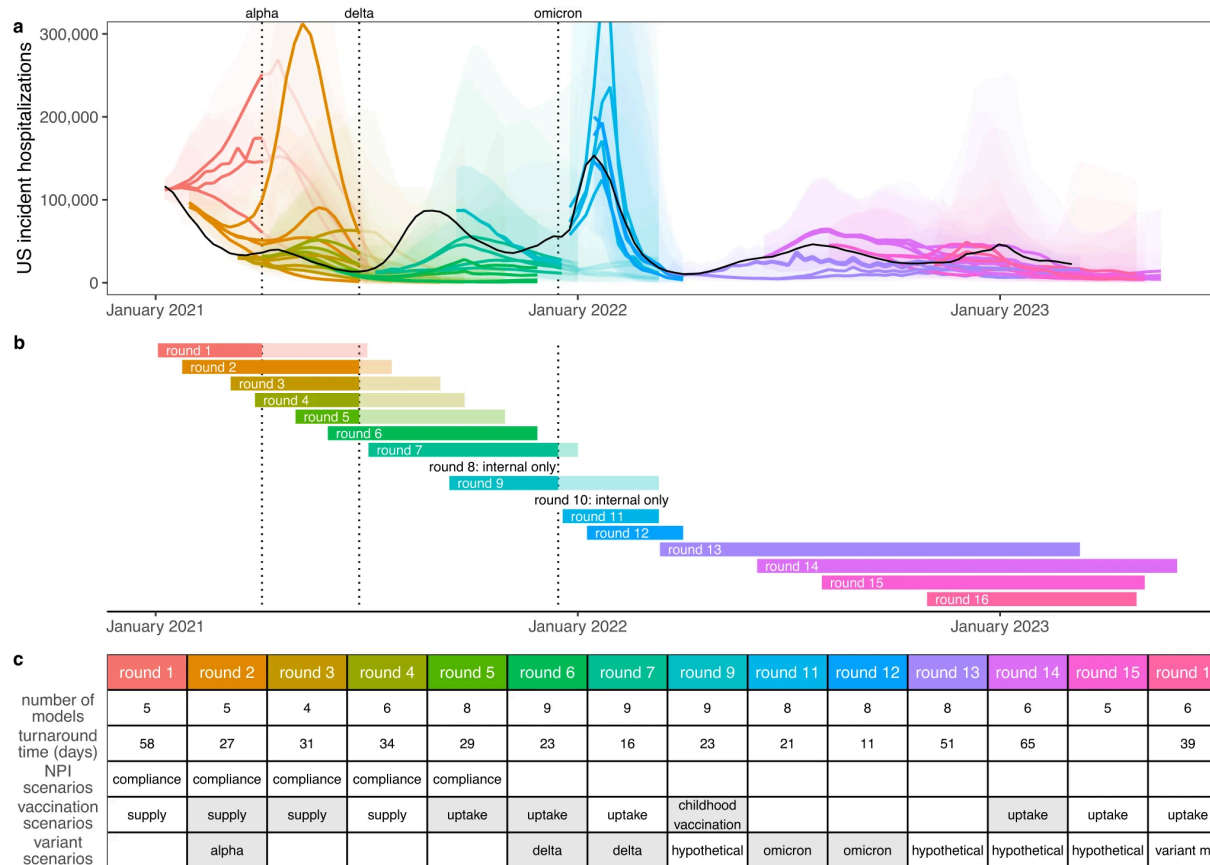
Humans are better than models at predicting *cases*, but not *deaths*



Bosse et al., *PLOS Comp Biol*, 2022

Broader evaluations: not just forecasts

Any model of the future is a prediction and can be evaluated as such.



Howerton et al., *Nat Comm*, 2023

Which models are useful?

Usefulness doesn't have to coincide with accuracy

offered by Kenneth Arrow. During the Second World War he was a weather officer in the US Army Air Corps working on the production of month-ahead weather forecasts, and this is how he tells the story [74]:

The statisticians among us subjected these forecasts to verification and they differed in no way from chance. The forecasters themselves were convinced and requested that the forecasts be discontinued. The reply read approximately like this: “The commanding general is well aware that the forecasts are no good. However, he needs them for planning purposes”.

Saltelli, *arXiv*, 2018

Assessing utility is not usually a priority



RESEARCH ARTICLE

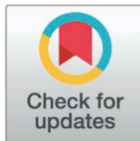
The utility of infectious disease modelling in informing decisions for outbreak response: A scoping review

Duaa Rao^{1,2}, Aleena Tanveer^{1,2}, Emil Nafis Iftekhar^{1*}, Sophie Alice Müller¹, Katharine Sherratt³, Klara Röbl¹, Paola Carrillo-Bustamante¹, Katharina Heldt⁴, Julia Fitzner⁵, Johanna Hanefeld^{1,6}, Sebastian Funk^{1,3}

1 Centre for International Health Protection, Robert Koch Institute, Berlin, Germany, **2** Institute of International Health, Charité—Universitätsmedizin Berlin, Berlin, Germany, **3** Centre for Mathematical Modelling of Infectious Diseases, London School of Hygiene & Tropical Medicine, London, United Kingdom, **4** Method Development, Research Infrastructure and Information Technology, Robert Koch Institute, Berlin, Germany, **5** World Health Organization Hub for Pandemic and Epidemic Intelligence, Berlin, Germany, **6** Department of Global Health and Development, London School of Hygiene & Tropical Medicine, London, United Kingdom

☯ These authors contributed equally to this work.

* lftekharE@rki.de



Rao et al., *medRxiv*, 2025

Back to the cruise ship



Photo: Stefan Brending

What if we were wrong?

"All models are wrong, but some are useful."

George Box

What if we were wrong?

"Remember that all models are wrong; the practical question is how wrong do they have to be to not be useful"

George Box

Thank you

Slides at



https://epiforecasts.io/slides/monid_20260521.html